



WELCOME!

Online Training Course
**Course 2: Foundations in Mobile Phone Data (MPD) for Policy:
Modern Data Workflows**

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Course 2: Foundations in Mobile Phone Data (MPD) for Policy: Modern Data Workflows Overview

Eight online training sessions (live webinars):

- Designing mobile phone data for policy pipelines
- **Privacy-by-design**
- Input data quality assurance and cleaning
- Continuity models
- Pols/Home Locations / Usual Environment
- Calculating "foundational" / "core" aggregates
- Quality Assessment of Estimates for Decision-Making
- From Aggregates to Estimates - population inference from mobile phone data and standard data sources

Five self-paced topics (on OLC):

- Recap of fundamentals of mobile phone data, data pipelines, data products, data types
- "Introduction to the course and examples use cases: data pipeline and stages
- Skill sets for each stage. Objectives."
- Mobile Phone Data: datasets, fields, types (links with 1.5)
- Reference and Complementary data etc.
- Considerations for multi-operators data pipelines

Housekeeping information

- Delivery timeslot for online training sessions (live webinars):
 - Every Wednesday until Wednesday 10 December
 - From **1pm-2pm UTC**
- Course 1 will only be available until end of October
- The slides and a session recording will be uploaded to the Open Learning Campus after delivery, for your reference
- Additional resources, quizzes and/or other follow-up will also be made available after sessions
- Try to keep Q&A until the end



Privacy-by-design for MPD pipelines

Lesson 2.3.3

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Introduction

This session is about the privacy-by-design approach, and how to build data governance into data pipelines. We will cover:

- The 7 principles of privacy by design
- The data governance journey, from identifying requirements to designing and then operating the pipeline, and what happens if something goes wrong
- The stages of privacy preservation during the data lifecycle
- Anonymisation techniques, including a brief overview of some emerging technologies

Learning objectives

- Describe why location data, including MPD, is difficult to anonymise
- Explain the concept of privacy-by-design
- Describe the common stages of privacy-preservation in an MPD pipeline
- Identify best practices for individual privacy-preservation and data security
- Identify data governance processes which should be built into an MPD pipeline

Contents

1. Recap of pre-reading material
2. Building data governance into MPD-for-policy pipelines
3. Stages of privacy preservation
4. Summary

Recap of pre-reading material

Location data

Subscriber ID	Timestamp	Location
1e4b0c6f	2025-10-29 11:09:21	...
1e4b0c6f	2025-10-29 13:32:19	...
1e4b0c6f	2025-10-29 13:54:06	...
1e4b0c6f	2025-10-29 19:37:45	...
ae7c98bd	2025-10-29 08:11:43	...

Is this dataset anonymous?

Why is location data difficult to anonymise?

- Individual location trajectories are **very unique**
 - Even without direct identifiers
 - Even in large datasets and at coarse resolution
- Individual location data is classified as **personal data** under GDPR, and usually as **special-category** personal data (e.g. if it reveals visits to hospitals, places of worship etc.)
- Vital to properly **anonymise** MPD-derived indicators before release, e.g. through aggregation and privacy-enhancing technologies (PETs)
- But anonymisation is a **moving target** - new methods developed over time, so a dataset that's anonymous today may not remain so forever

What is Data Governance?

Policies, processes and standards used to manage data assets effectively and responsibly - balance **data protection** with **usability**.

- Ownership and accountability
- Quality standards
- Access and security
- Compliance
- Data lifecycle management
- Documentation

Privacy by design

Privacy considerations embedded into systems and processes from the start.

1. Proactive not reactive; preventive not remedial
2. Privacy as the default setting
3. Privacy embedded into design
4. Full functionality - positive-sum, not zero-sum
5. End-to-end security - full lifecycle protection
6. Visibility and transparency - keep it open
7. Respect for user privacy - keep it user-centric

Building data governance into MPD for Policy pipelines

The data governance journey

- Identifying requirements
- Planning
- The pipeline in operation
- What if it goes wrong?

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Identifying requirements

- Specify the use case(s)
 - What outputs will we produce?
 - Who will use them, and why?
- Identify inputs
 - What data do we need?
 - How sensitive is it?
- Compliance requirements
 - Legal requirements (e.g. Malabo Convention, GDPR, national regulations)
 - Industry regulations (e.g. GSMA guidelines)
 - Company policies (e.g. MNO data handling policies)
 - E.g. consent requirements, retention/deletion durations, data transfer restrictions

The data governance journey

- Identifying requirements
- Planning
- The pipeline in operation
- What if it goes wrong?

Planning - risk assessment

Evaluate before you build.

- Is processing necessary and proportional?
 - Is all data collection strictly necessary for the specified use case?
- What are the privacy risks?
 - Re-identification
 - Data breach
 - Outputs used for unintended purposes
- How do we mitigate risks?
 - **Put answers in the chat**

Planning - risk assessment

Evaluate before you build.

- Is processing necessary and proportional?
 - Is all data collection strictly necessary for the specified use case?
- What are the privacy risks?
 - Re-identification
 - Data breach
 - Outputs used for unintended purposes
- How do we mitigate risks?
 - E.g. encryption, access control, policies, training
- Regulations (e.g. GDPR) may mandate a thorough data protection impact assessment (DPIA)

Planning - data classification

Know what data you have.

- How sensitive is each dataset (input, intermediate, output)?
- What restrictions are required (access control, retention/deletion, location of storage/processing)?

Possible example:

- MPD traces must be processed in-country
- Personal data accessed only by MNO staff
- Anonymised outputs to be encrypted; accessible to approved third parties only

Planning - basis of processing

Legal foundation for data use.

- Purpose of data processing
- Consent requirements
- Retention requirements

Possible example:

- No explicit consent required (covered by public interest), but data subjects must be given opportunity to object to data processing
- Collected data must only be used for the specified purpose - additional new use cases would require new legal agreements and consultation with data subjects
- Personal data to be retained for 5 years after processing, then erased

Planning - roles, policies, procedures

Put rules and processes in place.

- Roles (data ownership, pipeline maintenance, incident review, ...)
 - Responsibilities are clearly defined
 - Role assignment/membership is recorded and regularly reviewed
- Policies
 - Access permissions for each role
 - Data labelling
- Procedures
 - Backup/recovery
 - Incident response
 - Data retention/erasure
 - Pipeline deployment/update

The data governance journey

- Identifying requirements
- Planning
- The pipeline in operation
- What if it goes wrong?

The pipeline in operation

What happens while data is flowing through the pipeline?

- Automated compliance
 - Data labelling/classification
 - Retention/erasure
- Security
 - Role-based access control
 - Server hardening
 - Encryption
- Data quality checks and metrics
- Audit logging - keeping a paper trail
 - Who did what, when?

The data governance journey

- Identifying requirements
- Planning
- The pipeline in operation
- What if it goes wrong?

What if it goes wrong?

Things do go wrong - how do we protect against them?

→ **Put answers in the chat**

What if it goes wrong?

Things do go wrong - how do we protect against them?

- Breach detection
 - Early warning system - detect issues as soon as possible
 - Uses compliance/quality checks and audit logs
- Incident response
 - Roles and procedures were defined in the planning stage - follow them
- Continuous review
 - Learn from mistakes/problems, and adapt to changes

Stages of privacy preservation

Stages of privacy preservation

1. Data collection
2. Pseudonymisation
3. Secure storage and access control
4. Anonymisation
5. Retention and disposal

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Data collection

- Purpose limitation
 - Collect and process data only for a well-defined purpose
- Consent
 - Ensure data subjects give informed consent and/or can opt out
- Data minimisation
 - Do not collect any more than the necessary data

Stages of privacy preservation

1. Data collection
2. Pseudonymisation
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Pseudonymisation

- Replace direct identifiers (names, phone numbers, ...) with artificial identifiers
 - E.g. hashing
- *Not* the same as anonymisation (see earlier points - location data is personal data even without direct identifiers)
- Falls under data minimisation - for population-level outputs, individual identifiers are not required
 - If individual-level linkage to other datasets is required, those must be accordingly pseudonymised
- Data should be pseudonymised by the data controller (MNO) **before** any other party can access the data

Pseudonymisation

MSISDN	Timestamp	Cell ID
0123456789	2025-10-29 11:09:21	10925
0123456789	2025-10-29 13:32:19	10776
0987654321	2025-10-29 08:11:43	37422
0987654321	2025-10-29 13:54:06	12832
0246813579	2025-10-29 19:37:45	25956



MSISDN	Timestamp	Cell ID
1e4b0c6f	2025-10-29 11:09:21	10925
1e4b0c6f	2025-10-29 13:32:19	10776
641adba	2025-10-29 08:11:43	37422
641adba	2025-10-29 13:54:06	12832
ae7c98bd	2025-10-29 19:37:45	25956

Stages of privacy preservation

1. Data collection
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Secure storage and access control

- Encryption at rest
 - Note: trade-off between security and processing speed
- Access control (principle of least privilege)
 - Linked to pre-defined roles, and documented
- Server hardening
 - E.g. security updates, password policies, firewall, secrets rotation, penetration testing
- Data classification
 - Clearly-defined sensitivity levels, with corresponding handling policies
- Encryption in transit - secure transfer protocols (e.g. TLS/SSL)
 - Over private networks where possible
- Audit logging
 - And ensure system is monitored

Stages of privacy preservation

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Anonymisation

Individuals should not be identifiable in output data. Achieve this through:

- **Aggregation** - summarise data per group
 - E.g. number of residents per location, instead of home location per resident
- **Redaction** - hide/exclude sensitive subsets of data
 - E.g. data points relating to distinctive individuals or small groups
- **Statistical disclosure control (SDC)** - techniques include:
 - **Suppression** - hide data fields that could reveal identities
 - **Generalisation** - broaden categories (e.g. age bands instead of exact ages)
 - **Top/bottom coding** - group extreme values together (e.g. “age 85 and over”)
 - **Perturbation** - add small amount of noise to data

Anonymisation

k-anonymity: ensure every individual is indistinguishable from at least $k-1$ other individuals.

Using suppression ($k=2$):

Name (pseudonymised)	Age	Gender
1e4b0c6f	30	Female
641adba	24	Female
ee7f44b4	43	Male
ae7c98bd	49	Male



Name (pseudonymised)	Age	Gender
1e4b0c6f	*	Female
641adba	*	Female
ee7f44b4	*	Male
ae7c98bd	*	Male

Anonymisation

k-anonymity: ensure every individual is indistinguishable from at least $k-1$ other individuals.

Using generalisation ($k=2$):

Name (pseudonymised)	Age	Gender
1e4b0c6f	30	Female
641adba	24	Female
ee7f44b4	43	Male
ae7c98bd	49	Male



Name (pseudonymised)	Age	Gender
1e4b0c6f	20 < age ≤ 30	Female
641adba	20 < age ≤ 30	Female
ee7f44b4	40 < age ≤ 50	Male
ae7c98bd	40 < age ≤ 50	Male

Anonymisation

Similar for aggregated data (ensure all groups represent at least k individuals).

Using redaction (k=20):

Location	Resident count
District 1 - town A	1028
District 1 - town B	173
District 2 - town C	695
District 3 - town D	77
District 3 - town E	12



Location	Resident count
District 1 - town A	1028
District 1 - town B	173
District 2 - town C	695
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Anonymisation

Similar for aggregated data (ensure all groups represent at least k individuals).

Using generalisation (k=20):

Location	Resident count
District 1 - town A	1028
District 1 - town B	173
District 2 - town C	695
District 3 - town D	77
District 3 - town E	12



Location	Resident count
District 1	1201
District 2	695
District 3	89

Privacy-Enhancing Technologies (PETs)

Some more advanced PETs:

- **Differential privacy** - add random noise to mask individual contributions while preserving overall patterns
- **Federated learning** - train a model on decentralised devices (take the model to the data, not the data to the model)
- **Homomorphic encryption** - perform computations on encrypted data, such that the output can be decrypted to find the answer
- **Secure multi-party computation** - multiple parties jointly process their private data, without revealing the inputs to each other
 - Distributed cryptographic protocols
 - And/or use secure enclaves / trusted execution environments
- **Synthetic data** - produce a fake dataset that reproduces patterns from the real data

See the [UN PET Guide](#) for more details.

Stages of privacy preservation

1. Data collection
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3. Secure storage and access control
4. Anonymisation
5. Retention and disposal

Retention and disposal

- Secure backups
 - Ensure there is a backup of the data if the primary copy is lost
 - Backup must be secured, as with the primary copy
- Automated retention policies
 - Data that must be retained is securely stored
 - Erased automatically when finished
- Deletion requests
 - Ensure deletion requests from data subjects can be enacted efficiently
- Documentation, data classification and data lineage are key to enable this

Summary

Summary

- Location trajectory data is sensitive and hard to anonymise
- Privacy by design is an approach to embed privacy considerations throughout processes
- Data governance must be taken into account throughout pipeline implementation, from initial planning to deployment and ongoing maintenance
 - And throughout the data lifecycle, from data collection to erasure
- Outputs should be anonymised, e.g. using aggregation, k-anonymisation, and/or more advanced privacy-enhancing technologies
- Well-defined policies and roles, along with comprehensive logging and monitoring, are key to protecting sensitive data

Further reading

Further reading

Handbook on MPD for Official Statistics - UN Global Working Group on Big Data for Official Statistics (2019):

<https://unstats.un.org/bigdata/task-teams/mobile-phone/MPD%20Handbook%2020191004.pdf>

Guiding principles to maintain public trust in the use of mobile operator data for policy purposes - Jansen, R., Kovacs, K., Esko, S., Saluveer, E., Sõstra, K., Bengtsson, L., Li, T., Adewole, W., Nester, J., Arai, A., Magpantay, E. (2021). [doi:10.1017/dap.2021.21](https://doi.org/10.1017/dap.2021.21)

UN Guide on PETs for Official Statistics (2023): [https://unstats.un.org/bigdata/task-teams/privacy/guide/2023 UN%20PET%20Guide.pdf](https://unstats.un.org/bigdata/task-teams/privacy/guide/2023_UN%20PET%20Guide.pdf)



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